

PROJECT 2: EXPLORATORY DATA ANALYSIS (EDA)

Dataset Overview

- Dataset Name: Product Sales Region
- Total Records: 1500
- Total Columns: 19

1. Descriptive Statistics

Descriptive statistics were calculated for the key numerical variables including Quantity, TotalPrice, UnitPrice, Discount, and ShippingCost.

A	B	C	D	E	F
Metric	Quantity	TotalPrice	UnitPrice	Discount	ShippingCost
Mean	10.41067	3151.929	298.8269	0.073133	27.5072933
Median	11	2348.315	294.74	0.05	27.1
Count	1500	1500	1500	1500	1500
Min	1	8.2	5.52	0	5.01
Max	20	11526	599.72	0.15	49.98
StandardDeviation		2712.312			

Observation:

The dataset contains a balanced distribution of sales transactions. The difference between the mean and median of TotalPrice indicates the presence of a few high-value transactions.

2. Sales by Region Analysis

Sales by Region	
Region	Sum of TotalPrice
Central	914786.98
East	947363.41
North	1051081.78
South	893128.87
West	921531.9
Grand Total	4727892.94

Observation:

- North generated the highest sales revenue.
- South generated the lowest sales revenue.
- Overall sales were relatively balanced across all regions.

3. Product Performance Analysis

Product Performance	
Product	Sum of TotalPrice
Chair	669379.72
Desk	602045.39
Laptop	737135.01
Monitor	703619.5
Phone	536612.88
Printer	743505.81
Tablet	735594.63
Grand Total	4727892.94

Observation:

- Printer generated the highest sales revenue.
- Phone generated the lowest sales revenue.

4. Customer Type Analysis

Customer Type Analysis	
CustomerType	Count of OrderID
Retail	744
Wholesale	756
Grand Total	1500

Observation:

- Wholesale customers placed slightly more orders than Retail customers.

5. Payment Method Analysis

Payment Method Analysis	
PaymentMet	Count of OrderID
Cash	312
Credit Card	299
Debit Card	290
Gift Card	276
Online	323
Grand Total	1500

Observation:

- Online payment was the most preferred payment method.
- Gift Card was the least used payment method.

6. Returns Analysis

Returns Analysis	
Returned	Count of OrderID
0	1128
1	372
Grand Total	1500

Observation:

- Most orders were completed successfully.
- The return rate was relatively low compared to the total number of transactions.

7. Promotion Analysis

Promotion Analysis	
Promotion	Sum of TotalPrice
FREESHIP	1331296.19
No Promotion	1163832.31
SAVE10	1070452.23
WINTER15	1162312.21
Grand Total	4727892.94

Observation:

- FREESHIP generated the highest sales revenue.
- SAVE10 generated the lowest sales revenue.

8. Outlier Analysis

Outlier detection was performed on the TotalPrice column using descriptive statistics and Z-score analysis. The mean value of TotalPrice was 3151.93, while the maximum transaction value was 11526. The large difference between the average and maximum values indicates the presence of unusually high-value transactions.

Observation:

- A small number of transactions exhibited unusually high sales values.
- These transactions were identified as potential outliers.
- No records were removed because the values appeared to be valid business